



1 Nonproprietary Names

None adopted.

2 Synonyms

See Table I.

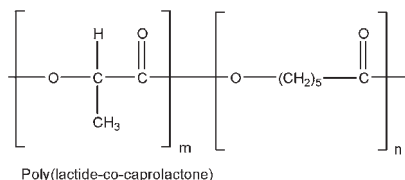
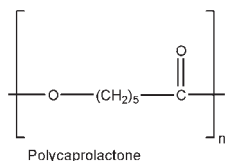
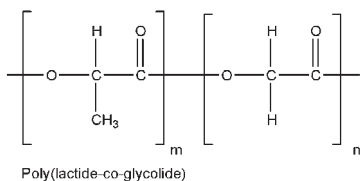
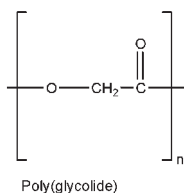
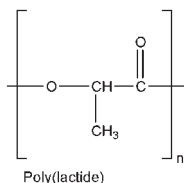
3 Chemical Name and CAS Registry Number

See Table I.

4 Empirical Formula and Molecular Weight

Aliphatic polyesters are synthetic homopolymers or copolymers of lactic acid, glycolic acid, lactide, glycolide and ϵ -hydroxycaproic acid. Typically, the molecular weights of homopolymers and copolymers range from 2000 to >100 000 Da.

5 Structural Formula



6 Functional Category

Bioabsorbable material; biocompatible material; biodegradable material.

7 Applications in Pharmaceutical Formulation or Technology

Owing to their reputation as safe materials and their biodegradability, aliphatic polyesters are primarily used as biocompatible and biodegradable carriers in many types of implantable or injectable drug-delivery systems for both human and veterinary use. Examples of implantable drug delivery systems include rods,⁽¹⁾ cylinders, tubing, films,⁽²⁾ fibers, pellets, and beads.⁽³⁾ Examples of injectable drug-delivery systems include microcapsules,⁽⁴⁾ microspheres,⁽⁵⁾ nanoparticles, and liquid injectable controlled-release systems such as gel formulations.⁽⁶⁾

8 Description

Aliphatic polyesters are a group of synthesized homopolymers or copolymers. They are nontoxic and can easily be fabricated into a variety of novel devices, such as rods, screws, nails, and cylinders. The polymers are commercially available in varying molecular weights as both homopolymers and copolymers. Molecular weights of polyesters range from 2000 Da to greater than 100 000 Da.

Co-monomer ratios of lactic acid and glycolic acid (or lactide and glycolide) for poly(DL-lactide-co-glycolide) range from 85:15 to 50:50. Table I shows the chemical and trade names of different commercially available aliphatic polyesters.

9 Pharmacopeial Specifications

—

10 Typical Properties

For typical physical and mechanical properties of the aliphatic polyesters, see Tables II, III, IV, V, VI, and VII.

Polymer composition and crystallinity play important roles in the solubility of these aliphatic polyesters. The crystalline homopolymers of glycolide or glycolic acid are soluble only in strong solvents, such as hexafluoroisopropanol. The crystalline homopolymers of lactide or lactic acid also do not have good solubility in most organic solvents. However, amorphous polymers of DL-lactide or DL-lactic acid and copolymers of lactide or lactic acid with a low glycolide or glycolic acid content are soluble in many organic solvents (Table II). Aliphatic polyesters are slightly soluble or insoluble in water, methanol, ethylene glycol, heptane, and hexane.

11 Stability and Storage Conditions

The aliphatic polyesters are easily susceptible to hydrolysis in the presence of moisture. Hence, they should be packaged under high-purity dry nitrogen and properly stored in airtight containers, preferably refrigerated at below 0°C. It is necessary to allow the polymers to reach room temperature in a dry environment before opening the container. After the original package has been opened, it is recommended to re-purge the package with high-purity dry nitrogen prior to resealing.

12 Incompatibilities

—

13 Method of Manufacture

Generally, aliphatic polyesters can be synthesized via polycondensation of hydroxycarboxylic acids and catalytic ring-opening polymerization of lactones. Ring-opening polymerization is preferred because polyesters with high molecular weights can be

Table I: Chemical names and CAS registry numbers of the aliphatic polyesters.

Generic name	Composition (%)			Synonyms	Trade name	Manufacturer	CAS name	CAS number
	Lactide	Glycolide	Caprolactone					
Poly(L-lactide)	100	0	0	L-PLA	<i>Lactel</i> L-PLA 100 L <i>Resomer</i> L 206 S, 207 S, 209 S, 210, 210 S	Durect Lakeshore BI	Poly[oxy[(1S)-1-methyl-2-oxo-1,2-ethanediy]]	[26161-42-2]
Poly(DL-lactide)	100	0	0	DL-PLA	<i>Lactel</i> DL-PLA <i>Purasorb</i> PDL 02A, 02, 04, 05 <i>Resomer</i> R 202 S, 202 H, 203 S, 203 H 100 DL 7E <i>Resomer</i> LG 855 S, 857 S	Durect Purac BI Lakeshore BI	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, homopolymer	[26680-10-4]
Poly(L-lactide-co-glycolide)	85	15	0		<i>Resomer</i> LG 824 S	BI	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, (3S,6S)-, polymer with 1,4-dioxane-2,5-dione	[30846-39-0]
Poly(L-lactide-co-glycolide)	82	18	0		<i>Resomer</i> GL 903	BI	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, (3S,6S)-, polymer with 1,4-dioxane-2,5-dione	[30846-39-0]
Poly(L-lactide-co-glycolide)	10	90	0			BI	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, (3S,6S)-, polymer with 1,4-dioxane-2,5-dione	[30846-39-0]
Poly(DL-lactide-co-glycolide)	85	15	0	Polyglactin;DL-PLGA (85 : 15)	<i>Lactel</i> 85 : 15 DL-PLG 8515 DLG 7E <i>Resomer</i> RG 858 S	Durect Lakeshore BI	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, polymer with 1,4-dioxane-2,5-dione	[26780-50-7]
Poly(DL-lactide-co-glycolide)	75	25	0	Polyglactin;DL-PLGA (75 : 25)	<i>Lactel</i> 75:25 DL-PLG <i>Purasorb</i> PDLG 7502A, 7502, 7507 <i>Resomer</i> RG 752 H, 752 S, 753 S, 755 S, 756 S 7525 DLG 7E	Durect Purac BI Lakeshore Durect Lakeshore BI	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, polymer with 1,4-dioxane-2,5-dione	[26780-50-7]
Poly(DL-lactide-co-glycolide)	65	35	0	Polyglactin;DL-PLGA (65 : 35)	<i>Lactel</i> 65:35 DL-PLG 6535 DLG 7E <i>Resomer</i> RG 653 H	Durect Lakeshore BI	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, polymer with 1,4-dioxane-2,5-dione	[26780-50-7]
Poly(DL-lactide-co-glycolide)	50	50	0	Polyglactin;DL-PLGA (50 : 50)	<i>Lactel</i> 50:50 DL-PLG 5050 DLG 7E, 5E, 1A, 2A, 3A, 4A, 4.5A <i>Purasorb</i> PDLG 5002A, 5002, 5004A, 5004, 5010 <i>Resomer</i> RG 502, 502H, 503, 503H, 504, 504H, 509S	Durect Lakeshore Purac BI	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, polymer with 1,4-dioxane-2,5-dione	[26780-50-7]
Poly-ε-caprolactone	0	0	100	PCL	<i>Lactel</i> PCL 100 PCL 8515 DL/PCL	Durect Lakeshore Lakeshore	2-Oxepanone, homopolymer	[24980-41-4]
Poly(DL-lactide-co-ε-caprolactone)	85	0	15		<i>Lactel</i> 80 : 20 DL-PLCL	Durect	1,4-Dioxane-2,5-dione,3,6-dimethyl-, polymer with 2-oxepanone	[70524-20-8]
Poly(DL-lactide-co-ε-caprolactone)	80	0	20	DL-PLCL (80 : 20)	<i>Lactel</i> 25 : 75 DL-PLCL	Durect	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, polymer with 2-oxepanone	[70524-20-8]
Poly(DL-lactide-co-ε-caprolactone)	25	0	75	DL-PLCL (25 : 75)	<i>Resomer</i> LC 703 S	BI	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, (3S,6S)-, polymer with 2-oxepanone	[65408-67-5]
Poly(L-lactide-co-ε-caprolactone)	70	0	30		8515 L/PCL	Lakeshore	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, (3S,6S)-, polymer with 2-oxepanone	[65408-67-5]
Poly(L-lactide-co-ε-caprolactone)	85	0	15		7525 L/PCL	Lakeshore	1,4-Dioxane-2,5-dione, 3,6-dimethyl-, (3S,6S)-, polymer with 2-oxepanone	[65408-67-5]

(a) BI, Boehringer Ingelheim; Durect, Durect Corporation; Lakeshore, Lakeshore Biomaterials; Purac, Purac America.

Table II: Typical physical and mechanical properties of the aliphatic polyesters.^(a)

	50/50 DL-PLG	65/35 DL-PLG	75/25 DL-PLG	85/15 DL-PLG	DL-PLA	L-PLA	PGA	PCL
Molecular weight	40 000–100 000	40 000–100 000	40 000–100 000	40 000–100 000	40 000–100 000	>100 000	>100 000	80 000–150 000
Inherent viscosity (dL/g)	0.5–0.8 ^(b)	0.5–0.8 ^(b)	0.5–0.8 ^(c)	0.5–0.8 ^(c)	0.5–0.8 ^(c)	0.9–1.2 ^(c)	1.4–1.8 ^(b)	1.0–1.3 ^(c)
Melting point (°C)	Amorphous	Amorphous	Amorphous	Amorphous	Amorphous	173–178	225–230	58–63
Glass transition temperature (°C)	45–50	45–50	50–55	50–55	50–60	60–65	35–40	–65 to –60
Color	White to light gold	White to light gold	White to light gold	White to light gold	White	White	Light tan	White
Solubility (at 5% w/w) ^(d)	MeCl ₂ , THF, EtOAc, C ₃ H ₆ O, CHCl ₃ , HFIP	MeCl ₂ , THF, EtOAc, C ₃ H ₆ O, CHCl ₃ , HFIP	MeCl ₂ , THF, EtOAc, C ₃ H ₆ O, CHCl ₃ , HFIP	MeCl ₂ , THF, EtOAc, C ₃ H ₆ O, CHCl ₃ , HFIP	MeCl ₂ , THF, EtOAc, C ₃ H ₆ O, CHCl ₃ , HFIP	MeCl ₂ , CHCl ₃ , HFIP	HFIP	MeCl ₂ , CHCl ₃ , HFIP
Approx. resorption (months)	1–2	3–4	4–5	5–6	12–16	>24	6–12	>24
Specific gravity	1.34	1.30	1.30	1.27	1.25	1.24	1.53	1.11
Tensile strength (psi)	6000–8000	6000–8000	6000–8000	6000–8000	4000–6000	8000–12 000	10 000+	3000–5000
Elongation (%)	3–10	3–10	3–10	3–10	3–10	5–10	15–20	300–500
Modulus (psi)	2–4 × 10 ⁵	2–4 × 10 ⁵	2–4 × 10 ⁵	2–4 × 10 ⁵	2–4 × 10 ⁵	4–6 × 10 ⁵	1 × 10 ⁶	3–5 × 10 ⁴

Note: DL-PLG: DL-poly(lactide-co-glycolide); DL-PLA: DL-poly(lactide); L-PLA: L-poly(lactide); PGA: poly(glycolide); PCL: poly-ε-caprolactone.

(a) Specifications obtained from Durect.

(b) (HFIP) hexafluoroisopropanol.

(c) (CHCl₃) chloroform.

(d) Partial listing only: MeCl₂, methylene chloride; THF, tetrahydrofuran; EtOAc, ethyl acetate; HFIP, hexafluoroisopropanol; C₃H₆O, acetone.

Table III: General mechanical properties of selected aliphatic polyesters.^(a)

Property	Polymer			
	5050 DL	100 DL	100 L (Custom)	90/10 L/DL (Custom)
Composition	Poly (DL-lactide-co-glycolide) (50 : 50)	Poly (DL-lactide)	Poly (L-lactide)	Poly (L-lactide-co-DL-lactide)
Break stress (psi)	8296	6108	7323	7614
Strain at break (%)	5.2	5.0	5.5	5.2
Yield stress (psi)	8371	6666	7678	8414
Strain at yield (%)	5.1	3.7	4.9	4.5
Modulus (psi)	189 340	207 617	182 762	210 680

(a) Specification obtained from Lakeshore Biomaterials.

Table IV: Mechanical properties of poly(L-lactide/caprolactone).^(a)

Property	Poly (L-lactide/caprolactone) grade				
	50/50	75/25	85/15	90/10	95/05
Tensile strength (psi)					
At maximum	80	1488	3254	6232	6900
At 100%	79	400	1822	—	—
At 300%	44	950	2615	—	—
Elongation (%)					
To yield	>1000	>400	>6.4	8.1	1.6
To failure	>1000	>400	>500	8.1	1.6
Modulus (kpsi)	0.1	5.3	84	167	185
Shore D-hardness	5	52	87	91	95
Specific gravity	1.2	1.2	1.23	1.25	1.26
Compression molding temperature (°C)	73–130	130 ± 15	140 ± 10	165 ± 5	165 ± 5

(a) Specifications obtained from Lakeshore Biomaterials.

Table VI: Glass transition temperature and melting point of selected biodegradable polymers.^(a)

Polymer	Composition	Glass transition temperature (°C)	Melting point (°C)
100 PGA	Poly (glycolic acid)	35–40	225–230
100 L	Poly (L-lactide)	56–60	173–178
9010 G/L	Poly (L-lactide-co-glycolide) (10 : 90)	35–45	180–200
100 DL	Poly (DL-lactide)	50–55	Amorphous ^(b)
8515 DL/G	Poly (DL-lactide-co-glycolide) (85 : 15)	50–55	Amorphous ^(b)
7525 DL/G	Poly (DL-lactide-co-glycolide) (75 : 25)	48–53	Amorphous ^(b)
6335 DL/G	Poly (DL-lactide-co-glycolide) (65 : 35)	45–50	Amorphous ^(b)
5050 DL/G	Poly (DL-lactide-co-glycolide) (50 : 50)	43–48	Amorphous ^(b)
8515 DL/PCL	Poly (DL-lactide-co-caprolactone) (85 : 15)	20–25	Amorphous ^(b)
8515 L/PCL	Poly (L-lactide-co-caprolactone) (85 : 15)	20–25	Amorphous ^(b)
7525 L/PCL	Poly (L-lactide-co-caprolactone) (75 : 25)	13–20	Amorphous ^(b)
100 PCL	Poly (caprolactone)	(–60) – (–65)	60

(a) Specifications obtained from Lakeshore Biomaterials.

(b) Process temperature range: 140–160°C.

Table VII: Solubility of selected aliphatic polyesters.^(a)

Polymer	Solvent						
	Ethyl acetate	Methylene chloride	Chloroform	Acetone	Dimethyl formamide (DMF)	Tetrahydrofuran (THF)	Hexafluoro-isopropanol (HFIP)
Poly (L-lactide)	NS	S	S	NS	NS	NS	S
Poly (DL-lactide)	S	S	S	S	S	S	S
Poly (DL-lactide-co-glycolide) (85 : 15)	S	S	S	S	S	S	S
Poly (DL-lactide-co-glycolide) (75 : 25)	S	S	S	S	S	S	S
Poly (DL-lactide-co-glycolide) (65 : 35)	S	S	S	S	S	S	S
Poly (DL-lactide-co-glycolide) (50 : 50)	SS	S	S	SS	S	SS	S
Poly (caprolactone)	S	S	S	S	S	S	S
Poly (L-lactide-co-caprolactone) (75 : 25)	S	S	S	S	S	S	S
Poly (DL-lactide-co-caprolactone) (80 : 20)	S	S	S	S	S	S	S
Poly (glycolic acid)	NS	NS	NS	NS	NS	NS	S

(a) Specifications obtained from Lakeshore Biomaterials.

NS, not soluble; SS, slightly soluble (degree of solubility is dependent on molecular weight or inherent viscosity); S, soluble.

Table V: Mechanical properties of poly (DL-lactide/caprolactone).^(a)

Property	Poly(DL-lactide/caprolactone) grade				
	60/40	75/25	85/15	90/10	95/05
Tensile strength (psi)					
At maximum	65	1300	1555	4453	5493
At 100%	65	224	1555	—	—
At 300%	43	332	1041	—	—
Elongation (%)					
To yield	—	—	—	5.6	—
To failure	>400	>600	>500	5.6	7.2
Modulus (kpsi)	0.1	1.05	6.04	106	135
Shore D-hardness	0	42	79	88	95
Specific gravity	—	1.20	1.22	1.24	—
Compression molding temperature (°C)	—	82–140	82–140	82–140	120

(a) Specifications obtained from Lakeshore Biomaterials.

produced. Moreover, the dehydration of hydroxycarboxylic acids to form lactones does not have to be carried to a high degree of completion. Lactones can easily be purified owing to the differences of their physical and chemical properties from those of the corresponding hydroxycarboxylic acid. The esterification of the carboxylic acid end group makes polymers more hydrophobic, which decreases the hydrolytic degradation rate of the polymers in the presence of water or moisture.

14 Safety

Poly(lactic acid) or poly(lactide), poly(glycolic acid) or poly(glycolide), poly(lactic-co-glycolic acid) or poly(lactide-co-glycolide), and polycaprolactone are used in parenteral pharmaceutical formulations and are regarded as biodegradable, biocompatible, and bioabsorbable materials. Their biodegradation products are nontoxic, noncarcinogenic, and nonteratogenic. In general, these polyesters exhibit very little hazard.

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Contact with eyes, skin, and clothing, and breathing the dust of the polymers should be avoided. Aliphatic polyesters produce acid materials such as hydroxyacetic and/or lactic acid in the presence of moisture; thus, contact with materials that will react with acids, especially in moist conditions, should be avoided.

16 Regulatory Status

GRAS listed. Included in the Canadian List of Acceptable Non-medicinal Ingredients. Poly(lactide) and poly(lactide-co-glycolide) have been used in medical products and medical devices approved by the FDA.

17 Related Substances

Lactic acid.

18 Comments

Aliphatic polyesters are a group of synthesized, nontoxic, biodegradable polymers. In an aqueous environment, the polymer backbone undergoes hydrolytic degradation, through cleavage of the ester linkages, into nontoxic hydroxycarboxylic acids. Aliphatic polyesters are eventually metabolized to carbon dioxide and water, via the citric acid cycle.

The rate of biodegradation and drug-release characteristics from injectable drug-delivery systems formulated with the aliphatic polyesters can be controlled by changing the physicochemical properties of the polymers, such as crystallinity, hydrophobicity, monomer stereochemistry, copolymer ratio, end group, and

polymer molecular weight or by changing the porosity and geometry of the formulation.

Due to their ability to form complexes with heavy metal ions, aliphatic polyesters are added to skin-protective ointments.⁽⁷⁾

19 Specific References

- Shim IK *et al.* Healing of articular cartilage defects treated with a novel drug-releasing rod-type implant after microfracture surgery. *J Control Release* 2008; **129**: 187–191.
- Aviv M *et al.* Gentamicin-loaded bioresorbable films for prevention of bacterial infections associated with orthopedic implants. *J Biomed Mater Res A* 2007; **83**: 10–19.
- Wang G *et al.* The release of cefazolin and gentamicin from biodegradable PLA/PGA beads. *Int J Pharm* 2004; **273**: 203–212.
- Snider C *et al.* Microenvironment-controlled encapsulation (MiCE) process: effects of PLGA concentration, flow rate, and collection method on microcapsule size and morphology. *Pharm Res* 2008; **25**: 5–15.
- Giovagnoli S *et al.* Physicochemical characterization and release mechanism of a novel prednisone biodegradable microsphere formulation. *J Pharm Sci* 2008; **97**: 303–317.
- Zare M *et al.* Effect of additives on release profile of leuprolide acetate in an in situ forming controlled-release system: in vitro study. *J Appl Polym Sci* 2008; **107**: 3781–3787.
- Hoeffner EM *et al.*, ed. *Fiedler Encyclopedia of Excipients for Pharmaceuticals, Cosmetics and Related Areas*, 5th edn. Munich, Germany: Editio Cantor Verlag Aulendorf, 2002; 1270.

20 General References

- Alexis F. Factors affecting the degradation and drug-release mechanism of poly(lactic acid) and poly[(lactic acid)-co-(glycolic acid)]. *Polymer Int* 2005; **54**: 36–46.
- Allison SD. Effect of structural relaxation on the preparation and drug release behavior of poly(lactic-co-glycolic)acid microparticle drug delivery systems. *J Pharm Sci* 2008; **97**: 2022–2035.
- Boehringer Ingelheim. *Resomer*. <http://www.boehringer-ingelheim.com/corporate/ic/pharmachem/products/resomer.asp> (accessed 16 February 2009).
- Durect Corporation. *Lactel* absorbable polymers. http://www.durect.com/wt/durect/page_name/bp (accessed 16 February 2009).
- Giteau A *et al.* How to achieve sustained and complete protein release from PLGA-based microparticles? *Int J Pharm* 2008; **350**: 14–26.
- Johansen P *et al.* Development of synthetic biodegradable microparticulate vaccines: a roller coaster story. *Expert Rev Vaccines* 2007; **6**: 471–474.
- Jostel A, Shalet SM. Prospects for the development of long-acting formulations of human somatropin. *Treat Endocrinol* 2006; **5**: 139–145.
- Kulkarni RK *et al.* Biodegradable poly(lactic acid) polymers. *J Biomed Mater Res* 1971; **5**: 169–181.
- Lakeshore Biomaterials. <http://www.lakeshorebio.com/products.html> (accessed 16 February 2009).
- Lewis H. Controlled release of bioactive agents from lactide/glycolide polymers. Chasin M, Langer R, eds. *Biodegradable Polymers as Drug Delivery Systems*. New York: Marcel Dekker, 1990; 1–41.
- Li SM *et al.* Structure–property relationships in the case of the degradation of massive aliphatic poly-(α -hydroxy acids) in aqueous media, part 1: poly(DL-lactic acid). *J Mater Sci Mater Med* 1990; **1**: 123–130.
- Mohamed F, van der Walle CF. Engineering biodegradable polyester particles with specific drug targeting and drug release properties. *J Pharm Sci* 2008; **97**: 71–87.
- Nguyen TH *et al.* Erosion characteristics of catalyzed poly(orthoester) matrices. *J Control Release* 1987; **5**: 1–12.
- Olivier JC. Drug transport to brain with targeted nanoparticles. *NeuroRx* 2005; **2**: 108–119.
- Pitt CG *et al.* Sustained drug delivery systems II: factors affecting release rates from poly(ϵ -caprolactone) and related biodegradable polyesters. *J Pharm Sci* 1979; **68**: 1534–1538.
- Purac America. <http://www.purac.com> (accessed 16 February 2009).
- Reed AM, Gilding DK. Biodegradable polymers for use in surgery-poly(glycolic)/poly(lactic acid) homo and copolymers: 2. *In vitro* degradation. *Polymer* 1981; **22**: 494–498.
- Shah SS *et al.* Poly(glycolic acid-co-DL lactic acid): diffusion or degradation controlled drug delivery? *J Control Release* 1992; **18**: 261–270.
- Smith AW. Biofilms and antibiotic therapy: is there a role for combating bacterial resistance by the use of novel drug delivery systems? *Adv Drug Deliv Rev* 2005; **57**: 1539–1550.

Tamilvanan S *et al.* Manufacturing techniques and excipients used during the design of biodegradable polymer-based microspheres containing therapeutic peptide/protein for parenteral controlled drug delivery. *PDA J Pharm Sci Technol* 2008; 62: 125–154.

Vert M *et al.* New insights on the degradation of bioresorbable polymeric devices based on lactic and glycolic acids. *Clin Mater* 1992; 10: 3–8.

Visser GE *et al.* Biodegradation and tissue reaction to 50:50 poly(DL-lactide-co-glycolide) microcapsules. *J Biomed Mater Res* 1985; 19: 349–365.

Williams DF. Mechanisms of biodegradation of implantable polymers. *Clin Mater* 1992; 10: 9–12.

21 Authors

RK Chang, W Qu, AJ Shukla, N Trivedi.

22 Date of Revision

16 February 2009.

Alitame

1 Nonproprietary Names

None adopted.

2 Synonyms

Aclame; L-aspartyl-D-alanine-N-(2,2,4,4-tetramethylthietan-3-yl)amide; 3-(L-aspartyl-D-alaninamido)-2,2,4,4-tetramethylthietane.

3 Chemical Name and CAS Registry Number

L- α -Aspartyl-N-(2,2,4,4-tetramethyl-3-thietanyl)-D-alaninamide anhydrous [80863-62-3]

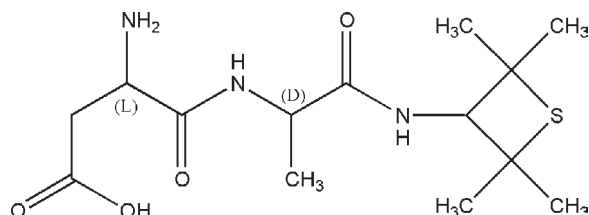
L- α -Aspartyl-N-(2,2,4,4-tetramethyl-3-thietanyl)-D-alaninamide hydrate [99016-42-9]

4 Empirical Formula and Molecular Weight

C₁₄H₂₅N₃O₄S 331.44 (for anhydrous)

C₁₄H₂₅N₃O₄S·2½H₂O 376.50 (for hydrate)

5 Structural Formula



6 Functional Category

Sweetening agent.

7 Applications in Pharmaceutical Formulation or Technology

Alitame is an intense sweetening agent developed in the early 1980s and is approximately 2000 times sweeter than sucrose. It has an insignificant energy contribution of 6 kJ (1.4 kcal) per gram of alitame.

Alitame is currently primarily used in a wide range of foods and beverages at a maximum level of 40–300 mg/kg.⁽¹⁾

8 Description

Alitame is a white nonhygroscopic crystalline powder; odorless or having a slight characteristic odor.

9 Pharmacopeial Specifications

—

10 Typical Properties

Acidity/alkalinity pH = 5–6 (5% w/v aqueous solution)

Isoelectric point pH 5.6

Melting point 136–147°C

Solubility see Table I.

Table I: Solubility of alitame.

Solvent	Solubility at 20°C unless otherwise stated
Chloroform	1 in 5000 at 25°C
Ethanol	1 in 1.6 at 25°C
<i>n</i> -Heptane	Practically insoluble
Methanol	1 in 2.4 at 25°C
Propylene glycol	1 in 1.9 at 25°C
Water	1 in 8.3 at 5°C
	1 in 7.6 at 25°C
	1 in 3.3 at 40°C
	1 in 2.0 at 50°C

Specific rotation $[\alpha]_D^{25} = +40^\circ$ to $+50^\circ$ (1% w/v aqueous solution)

11 Stability and Storage Conditions

Alitame is stable in dry, room temperature conditions but undergoes degradation at elevated temperatures or when in solution at low pH. Alitame can degrade in a one-stage process to aspartic acid and alanine amide (under harsh conditions) or in a slow two-stage process by first degrading to its β -aspartic isomer and then to aspartic acid and alanine amide. At pH 5–8, alitame solutions at 23°C have a half-life of approximately 4 years. At pH 2 and 23°C the half-life is 1 year.

Alitame should be stored in a well-closed container in a cool, dry place.

12 Incompatibilities

Alitame may be incompatible with oxidizing and reducing substances or strong acids and bases.

13 Method of Manufacture

Alitame may be synthesized by a number of routes.^(2,3) For example, 3-(D-alaninamido)-2,2,4,4-tetramethylthietane is dissolved in water and L-aspartic acid *N*-thiocarboxyanhydride is then added in portions with vigorous stirring, maintaining the pH of 8.5–9.5. The pH is then adjusted to 5.5 and *p*-toluenesulfonic acid